ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 1 style (theory). Section A: objective questions (30) · Section B: transistor amplifier biasing and configurations, op-amp amplifiers, ADC and DAC calculations, flip-flops, counters and shift registers, sensors and signal conditioning, SCRs, PWM and optocouplers (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places. Take V_{BE} = 0,7 V.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Voltage-divider bias is used in a BJT amplifier to …

A keep the operating point stable. B switch the transistor off.
C increase the noise. D remove the need for a supply.

1.1.2 The phase shift between the input and output of a common-emitter amplifier is …

A 0°. B 90°. C 180°. D 360°.

1.1.3 The voltage gain of a non-inverting op-amp amplifier is …

A $-R_f \div R_i$ B $1 + R_f \div R_i$ C $R_f \times R_i$ D $R_i \div R_f$

1.1.4 An 8-bit ADC divides its input range into … levels.

A 8 B 128 C 1 024 D 256

1.1.5 A flip-flop stores …

A one bit. B one byte.
C an analogue value. D a whole program.

1.1.6 A MOD-10 counter counts from …

A 0 to 9. B 0 to 10. C 1 to 9. D 0 to 15.

1.1.7 A strain gauge measures …

A temperature. B light. C strain (force). D sound.

1.1.8 The device used in AC lamp dimmers and motor-speed controllers is the …

A LED. B Zener diode. C triac. D thermistor.

1.1.9 The circuit that changes a digital number into an analogue voltage is a/an …

A ADC. B DAC. C comparator. D counter.

1.1.10 A MOSFET is controlled by its …

A base current. B gate voltage.
C light input only. D temperature.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 The output of an op-amp comparator switches between its two saturation levels.

1.2.2 A 10-bit ADC with a 5 V reference has a resolution of about 4,9 mV.

1.2.3 A shift register moves its data one position with each clock pulse.

1.2.4 A JK flip-flop with J = K = 1 keeps the same output on every clock pulse.

1.2.5 An emitter-follower (common-collector) amplifier has a high voltage gain.

1.2.6 A Wheatstone bridge can be used with a strain gauge.

1.2.7 On DC, an SCR switches off as soon as its gate signal is removed.

1.2.8 A thermocouple produces a voltage from a difference in temperature.

1.2.9 PWM controls the power to a motor efficiently.

1.2.10 Using more bits in an ADC makes each step larger.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	DC conditions that set an amplifier's quiescent (Q) point	A	counter
1.3.2	Op-amp circuit that adds several input voltages	B	optocoupler
1.3.3	Smallest input change that an ADC can detect	C	biasing
1.3.4	Op-amp circuit with a gain of 1, used as a buffer	D	low-pass filter
1.3.5	Circuit that counts clock pulses	E	voltage follower
1.3.6	Resistor whose value changes a lot with temperature	F	SCR
1.3.7	Bridge circuit that measures very small resistance changes	G	summing amplifier
1.3.8	Four-layer device used as a controlled rectifier	H	Wheatstone bridge
1.3.9	Isolates two circuits by passing the signal as light	I	resolution
1.3.10	Removes high-frequency noise from a sensor signal	J	thermistor

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: TRANSISTOR AMPLIFIERS

2.1 A common-emitter amplifier uses voltage-divider bias with V_{CC} = 12 V, R_1 = 47 kΩ, R_2 = 10 kΩ, R_E = 1 kΩ and R_C = 3,3 kΩ. Calculate:

2.1.1 The base voltage V_B (2)

2.1.2 The emitter voltage V_E (1)

2.1.3 The emitter current (take $I_C \approx I_E$) (2)

2.1.4 The collector voltage V_C (2)

2.2 State the phase relationship between the input and the output of this amplifier, and explain why coupling capacitors are used at the input and output. (3)

2.3 Give ONE typical use of EACH of the common-emitter, common-collector and common-base configurations. (3)

2.4 Explain what thermal runaway is and how the emitter resistor helps to prevent it. (3)

[16]

QUESTION 3: OP-AMPS AND DATA CONVERSION

3.1 A non-inverting amplifier has R_f = 90 kΩ and R_i = 10 kΩ. The input is 0,25 V. Calculate the gain and the output voltage. (3)

3.2 An inverting summing amplifier has R_f = 10 kΩ and two input resistors of 10 kΩ. The inputs are 1 V and 2 V. Calculate the output voltage. (3)

3.3 A 10-bit ADC uses a 5 V reference. Calculate:

3.3.1 The resolution (2)

3.3.2 The digital value for an input of 2 V (2)

3.4 Explain what is meant by the sampling rate of an ADC and why it must be high enough. (3)

3.5 A 4-bit DAC has a 5 V reference. Calculate the output voltage for the input 1011, using $V_{out} = (D \div 2^n) \times V_{ref}$. (3)

3.6 Explain why a voltage follower is often placed between a sensor and an ADC. (2)

[18]

QUESTION 4: DIGITAL SEQUENTIAL CIRCUITS

4.1 Draw the truth table of a clocked D flip-flop and explain its operation. (4)

4.2 Explain the difference between an asynchronous (ripple) counter and a synchronous counter. (4)

4.3 How many flip-flops are needed for a MOD-12 counter? Explain. (2)

4.4 A 4-bit shift register (Q3 Q2 Q1 Q0) holds 1011. It shifts right twice with a 0 entering at Q3. Write down the contents after EACH clock pulse. (3)

4.5 Give TWO uses of shift registers in robots. (2)

4.6 Explain what contact bounce is and how it can be removed. (3)

[18]

QUESTION 5: SENSORS, SIGNAL CONDITIONING AND POWER ELECTRONICS

5.1 Explain how a strain gauge in a Wheatstone bridge is used to measure force. (3)

5.2 Explain how an NTC thermistor in a voltage divider gives a voltage that changes with temperature. (2)

5.3 Name FOUR functions of signal conditioning. (4)

5.4 Explain how an SCR is triggered into conduction and how it switches off in an AC circuit. (3)

5.5 A motor driver applies PWM from a 12 V supply at 1 kHz with a duty cycle of 40%. Calculate the average voltage, the period and the on-time. (4)

5.6 State the purpose of an optocoupler between a microcontroller and a motor driver. (2)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER